CHAPTER 3

Hardware Architecture & Computer Systems

Processors, Memory Systems, Input/Output, Control Systems & Logic Circuits

LEARNING GOALS

In this chapter, you will learn about:

- Primary memory & secondary storage technologies
- RAM variants: SRAM vs. DRAM and usage cases
- Embedded systems, microcontrollers & trade-offs
- RISC vs. CISC processor design philosophies
- Parallel architectures: SISD, SIMD, MISD, MIMD
- Interrupt handling mechanisms on RISC & CISC
- 3D Printing methods: Direct vs. Binder 3D printing
- Boolean algebra expression simplification
- Hardware classification: Input, Output, Storage & CPU
- ROM types: Standard ROM, PROM, EPROM & EEPROM
- Sensors, actuators, ADCs, DACs & control feedback
- CPU registers, pipelining & execution hazards
- Supercomputers, clusters & massively parallel systems
- Display & touch technologies (Capacitive/Resistive, OLED)
- Logic gates (AND, OR, NOT, NAND, NOR, XOR) & truth tables
- AI Hardware acceleration (GPUs & Neural Engines)

KEY TERMS

Hardware: Physical components of a computer system.

Input Device: Enters data/instructions into a computer.

Output Device: Presents results/processed data to users.

Processing Device / CPU: Main circuit executing program instructions.

Primary Storage: Directly CPU-accessible temporary/working memory.

Secondary Storage: Non-volatile long-term data retention media.

Volatile Memory: Memory losing content when power is lost.

Non-Volatile Memory: Retains stored data without electrical power.

RAM / ROM: Random Access Memory / Read-Only Memory.

SRAM / DRAM: Static RAM (flip-flops) / Dynamic RAM (capacitors).

Refreshed: Periodic electrical recharging of DRAM cells.

PROM / EPROM: Programmable ROM / UV-Erasable Programmable ROM.

EEPROM / Flash: Electronically Erasable PROM / Block-based EEPROM.

HDD Latency: Delay while disk sector rotates to read-write head.

Fragmented: Non-consecutive storage of file sectors across a disk.

SSD: Solid-State Drive utilizing flash memory with no moving parts.

CISC / RISC: Complex / Reduced Instruction Set Computer.

Pipelining: Overlapping instruction execution stages in CPU.

Parallel Processing: Running multiple instruction streams concurrently.

Flynn's Taxonomy: SISD, SIMD, MISD, MIMD hardware classifications.

Cluster / Supercomputer: Networked SIMD/MIMD nodes / High-power mainframes.

Massively Parallel: Linking thousands of processors as a single engine.

ADC / DAC: Analogue-to-Digital / Digital-to-Analogue Converters.

Sensor / Actuator: Detects environmental data / Executes physical action.

Embedded System: Single-purpose computer inside a larger device.

Microcontroller: Integrated IC with CPU, RAM, ROM, and I/O pins.

3.1 Processing Architectures & CPU Design

The **Central Processing Unit (CPU)** is the computational core of a computer system. Its main role is to execute program instructions stored in memory by performing Fetch-Decode-Execute cycles. Computer architecture design dictates how efficiently instructions are processed and how hardware components communicate.

RISC vs. CISC Architectures

Processor instruction set architectures (ISAs) fall primarily into two design philosophies:

- **CISC (Complex Instruction Set Computer):** Designed to execute complex tasks in as few assembly lines as possible. CISC processors (like x86 architecture found in modern PCs) feature variable-length instructions and complex hardware decoding circuits. A single CISC instruction might perform a memory load, arithmetic operation, and memory store simultaneously.
- **RISC (Reduced Instruction Set Computer):** Designed with a small set of simple, fixed-length instructions that each execute in a single clock cycle. RISC architectures (like ARM processors found in mobile devices and modern supercomputers) use a load/store architecture where arithmetic operations occur strictly between internal registers.

Feature	CISC (Complex Instruction Set)	RISC (Reduced Instruction Set)
Instruction Size	Variable (1 byte to several bytes).	Fixed (typically 32-bit or 64-bit).
Execution Speed	Multiple clock cycles per instruction.	Single clock cycle per instruction.
Memory Access	Instructions operate directly on memory.	Strict Load/Store architecture (registers only).
Transistor Layout	Complex decoder circuits; fewer registers.	Simple decoders; large register file.
Pipelining Efficiency	Difficult due to variable instruction length.	Highly efficient due to fixed instruction sizes.

Instruction Pipelining and Processor Registers

Pipelining is a technique used in RISC processors to maximize instruction throughput. Instead of waiting for one instruction to complete all stages (Fetch, Decode, Execute, Writeback) before fetching the next, pipelining overlaps the stages of multiple instructions concurrently.

FIGURE 3.1 — 4-STAGE INSTRUCTION PIPELINE EXECUTION GRID

Clock Cycle	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5
Instruction 1	Fetch (F)	Decode (D)	Execute (E)	Writeback (W)	—
Instruction 2	—	Fetch (F)	Decode (D)	Execute (E)	Writeback (W)
Instruction 3	—	—	Fetch (F)	Decode (D)	Execute (E)

Pipeline Hazards: Pipelining can stall due to hazards:

- *Data Hazard:* An instruction depends on the unwritten result of a preceding instruction.
- *Control Hazard:* Branch instructions (e.g., conditional jumps) alter the execution flow, requiring pipeline flushing.
- *Structural Hazard:* Two instructions require the same hardware resource simultaneously.

Core CPU Registers: High-speed internal memory locations used during processing:

- **Program Counter (PC):** Holds the memory address of the next instruction to be fetched.
- **Memory Address Register (MAR):** Holds the RAM address currently being read from or written to.
- **Memory Data Register (MDR):** Stores data read from or waiting to be written to memory.
- **Current Instruction Register (CIR):** Holds the instruction currently being decoded and executed.
- **Accumulator (ACC):** Temporarily holds intermediate arithmetic and logic results from the ALU.

Parallel Computer Architectures (Flynn's Taxonomy)

To increase performance beyond single-core limits, parallel processing splits tasks across multiple execution units simultaneously. Michael J. Flynn classified computer architectures based on instruction and data streams:

FIGURE 3.2 — FLYNN'S ARCHITECTURAL CLASSIFICATION MATRIX

Architecture	Instruction Streams	Data Streams	Description & Common Application
SISD (Single Instruction Single Data)	Single	Single	Traditional single-core CPU executing one instruction on one data item at a time.
SIMD (Single Instruction Multiple Data)	Single	Multiple	Array processors / GPUs applying one operation simultaneously to multiple data arrays (e.g., image processing, vector math).
MISD (Multiple Instruction Single Data)	Multiple	Single	Rare architecture where multiple processors perform different operations on the same data stream (e.g., fault-tolerant space systems).
MIMD (Multiple Instruction Multiple Data)	Multiple	Multiple	Multi-core CPUs, supercomputers, and modern distributed systems where independent cores process different datasets independently.

Supercomputers, Clusters, and Massively Parallel Processing

When processing demands exceed a single machine's capability, specialized high-performance computing structures are used:

- **Clusters:** A collection of standalone computers (nodes) connected via high-speed local area networks, operating together as a unified resource. Nodes frequently employ SIMD or multi-core MIMD units.
- **Supercomputers:** Custom-built mainframes housing tens of thousands of processing cores sharing fast interconnects and memory pipelines, optimized for heavy scientific modeling.
- **Massively Parallel Computers (MPP):** Systems linking thousands of independent processors, each with its own memory, running programs concurrently via coordinated message-passing interfaces.

Interrupt Handling Mechanisms

An **Interrupt** is a signal generated by hardware or software requiring immediate CPU attention, pausing current program execution.

- **Handling Process:**
  1. CPU completes current instruction cycle.
  2. Contents of registers (PC, ACC, CIR) are pushed onto a memory stack.
  3. Interrupt Service Routine (ISR) address is loaded into PC to manage the event.
  4. Once resolved, saved register states are restored from stack and normal execution resumes.
- **CISC vs. RISC Handling:** CISC checks for interrupts between complex instruction cycles or multi-cycle steps, requiring large context saves. RISC handles interrupts swiftly due to uniform single-cycle instructions and fast register swapping.

Artificial Intelligence Hardware

Modern AI workloads (such as neural network training and inference) require massive linear algebra and matrix multiplication operations.

- **Graphics Processing Units (GPUs):** Specialized SIMD processors containing thousands of small efficiency cores designed to execute high-throughput mathematical operations in parallel.
- **Dedicated AI Accelerators (TPUs/NPUs):** Purpose-built integrated circuits tailored for low-precision tensor operations, maximizing power efficiency for deep learning models.

3.2 Primary Memory & Storage Technologies

Computer storage is structured hierarchically based on access speed, cost, capacity, and volatility. Storage is broadly divided into **Primary Storage** (directly accessible by the CPU) and **Secondary Storage** (non-volatile retention media).

Primary Storage: RAM vs. ROM

Primary memory communicates directly with the CPU over the system bus.

- **Random Access Memory (RAM):** Read-write, **volatile** primary memory that temporarily holds operating system files, active programs, and working data. Data is lost immediately when power supply ceases.
- **Read-Only Memory (ROM): Non-volatile** memory containing critical bootstrap instructions (BIOS/UEFI firmware) required to boot the system. It can be read from, but not easily written to during standard operation.

Random Access Memory Variants: SRAM vs. DRAM

Characteristic	SRAM (Static RAM)	DRAM (Dynamic RAM)
Storage Mechanism	Uses flip-flop logic circuits (4–6 transistors per bit).	Uses 1 transistor and 1 capacitor per bit.
Refresh Requirement	**No refresh needed** while powered.	**Requires periodic refresh** as capacitors leak charge.
Access Speed	Extremely fast (1–10 ns).	Slower than SRAM (30–60 ns).
Density & Cost	Lower bit density; high cost per MB.	Very high bit density; lower cost per GB.
Primary Application	CPU **Memory Cache** (L1, L2, L3 caches).	Main computer system RAM modules.

Read-Only Memory (ROM) Evolution

- **Standard ROM:** Pre-programmed during chip manufacturing with permanent binary masks; contents can never be changed.
- **PROM (Programmable ROM):** Manufactured blank. Can be programmed once by burning internal microscopic fuses using a high-voltage programmer.
- **EPROM (Erasable Programmable ROM):** Contains a transparent quartz window. Exposure to strong ultraviolet (UV) light clears all memory cells, allowing reprogramming.
- **EEPROM (Electronically Erasable PROM):** Contents erased and rewritten electronically using pulsed voltages without physical removal from the circuit board.
- **Flash Memory:** A specialized high-density form of EEPROM that reads/writes data in blocks, forming the foundation of modern solid-state storage.

Secondary Storage Media Categories

Secondary storage is non-volatile media used to store programs and user files permanently.

1. MAGNETIC STORAGE (HDDS)

Uses spinning magnetic platters read by electromagnetic heads.
• **Latency:** Rotational delay waiting for disk sector to pass under the head.
• **Fragmentation:** File sectors scattered in non-consecutive locations degrade read speeds.
• **Removable HDDs:** Portable external drives connected via USB for backups.

2. SOLID-STATE STORAGE (SSDS)

Uses NAND flash memory cells with floating-gate transistors.
• **No Moving Parts:** Zero latency, shock-resistant, highly durable.
• **High Performance:** Extremely fast read/write throughput via electronic grid access.

3. OPTICAL STORAGE (CDS, DVDS, BLU-RAY™)

Reads data using laser light reflected off reflective pits and lands stamped into polycarbonate discs.
• **Dual Layering:** DVDs and Blu-rays stack two reflective layers, nearly doubling storage capacity.
• **Birefringence:** A reading error where laser light refracts into two distinct beams inside imperfect disc substrate materials, causing data corruption.

3.3 Input, Output, Monitoring & Control Systems

Computer systems interact with the physical world through input devices, output devices, sensors, and actuators. Processing real-world phenomena requires translating continuous physical signals into discrete digital representations.

Signal Conversion: ADC and DAC

- **ADC (Analogue-to-Digital Converter):** Converts continuous physical signals (e.g., voltage, sound waves, temperature readings) into digital binary values understood by the CPU.
- **DAC (Digital-to-Analogue Converter):** Converts digital binary output values back into continuous electrical currents to drive physical actuators, motors, or speakers.

Sensors, Actuators, and Control Feedback Loops

- **Sensors (Input Devices):** Measure physical conditions (e.g., temperature, light level, pressure, humidity, motion, pH). They produce analogue signals sent to ADCs.
- **Actuators (Output Devices):** Electromechanical devices (e.g., motors, valves, relays, solenoids) that execute physical movement based on DAC control signals.
- **Monitoring System:** Reads sensor data continuously to display or log environment metrics without altering the environment directly.
- **Control System:** Reads sensor data, processes it against predefined threshold rules, and uses feedback loops to drive actuators to alter the environment.



Touchscreen Technologies

Touchscreens serve as combined input and display devices using two primary technologies:

Feature	Capacitive Touchscreens	Resistive Touchscreens
Operating Principle	Glass layers coated with conductive material store electric charge; fingers alter electrostatic field.	Flexible top plastic layer pushes against conductive bottom glass layer, completing circuit.
Multi-touch & Input	Supports multi-touch gestures; requires conductive input (bare finger or capacitive stylus).	Single touch focus; operated with any object (gloved hand, plastic stylus).
Durability & Clarity	High optical clarity; highly durable glass top.	Lower clarity; vulnerable to scratches; less expensive to manufacture.

Display Technologies & Advanced Peripherals

- **OLED (Organic LED):** Uses organic compounds that emit light when electric current passes through. Generates its own light, eliminating backlighting for true blacks, thinner screens, and higher energy efficiency.
- **Screen Resolution:** Number of distinct horizontal and vertical pixels (e.g., 1920 × 1080) displaying images.
- **Virtual Reality (VR) Headset:** Head-mounted apparatus featuring two separate displays, lenses, and motion sensors to totally immerse users in virtual surroundings.
- **3D Printing Technologies:**
  • *Direct 3D Printing:* Print head moves along X, Y, and Z axes ejecting melted polymer layers through nozzles.
  • *Binder 3D Printing:* Two-pass method using dry powder spread across a bed followed by a liquid binder spray.

Embedded Systems and Microcontrollers

An **Embedded System** is a specialized computer built into a larger machine to perform a specific dedicated control function (e.g., washing machine controller, ABS brake module). They typically use a **Microcontroller**—an integrated IC combining CPU core, RAM, ROM, and I/O ports on a single chip.

Benefits: Compact size, low power consumption, low per-unit cost, high operational reliability.
Drawbacks: Difficult to upgrade, specialized repair required, limited processing capacity.

3.4 Logic Circuits & Boolean Algebra

At the microscopic level, all computer hardware is constructed from digital logic gates that manipulate binary signals (0 and 1).

Fundamental Logic Gates

AND OPERATION (A · B)

Output is 1 ONLY if ALL inputs are 1.

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

OR OPERATION (A + B)

Output is 1 if AT LEAST ONE input is 1.

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

NOT OPERATION ($\bar{A}$)

Inverts the input signal.

A	Output
0	1
1	0

XOR OPERATION ($A \oplus B$)

Output is 1 if inputs are DIFFERENT.

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Boolean Algebra Simplification Laws

Boolean expressions describe logic circuits mathematically and can be simplified using formal laws:

- **Identity Laws:** $A \cdot 1 = A$ | $A + 0 = A$
- **Null Laws:** $A \cdot 0 = 0$ | $A + 1 = 1$
- **Idempotent Laws:** $A \cdot A = A$ | $A + A = A$
- **Inverse Laws:** $A \cdot \bar{A} = 0$ | $A + \bar{A} = 1$
- **Double Negation:** $\bar{\bar{A}} = A$
- **De Morgan's Laws:**
  - Break the line, change the sign: $\mathbf{A \cdot B = \bar{A} + \bar{B}}$
  - $\mathbf{A + B = \bar{A} \cdot \bar{B}}$

WORKED SIMPLIFICATION EXAMPLE

Simplify the Boolean Expression: $\mathbf{X = A \cdot B + A \cdot \bar{B}}$
1. Factor out term A using Distributive Law: $\mathbf{X = A \cdot (B + \bar{B})}$
2. Apply Inverse Law ($B + \bar{B} = 1$): $\mathbf{X = A \cdot (1)}$
3. Apply Identity Law: $\mathbf{X = A}$

End-of-Chapter Assessment Exercises

EXERCISE QUESTIONS

1. Storage Classification & Categories
a) Differentiate between primary storage and secondary storage in terms of volatility, access speed, and direct CPU accessibility. [3 marks]
b) Give one specific example of a secondary storage technology for each of the three main storage categories: magnetic, optical, and solid-state (flash). [3 marks]

2. Memory Architecture & Silicon Technologies
a) Explain the fundamental structural difference between SRAM (Static RAM) and DRAM (Dynamic RAM) regarding how data bits are physically stored and their requirement for periodic refreshing. [4 marks]
b) Contrast standard ROM (Read-Only Memory) with PROM (Programmable ROM) and EPROM (Erasable Programmable ROM) in terms of how data is written and erased. [4 marks]

3. Component Classification
Classify the following hardware components as Input, Output, or Storage devices (note dual roles where applicable):
i. Touchscreen display (consider both perspectives) [2 marks]
ii. Solid State Drive (SSD) [1 mark]
iii. Barcode reader [1 mark]

4. Automated Systems & Environment Control
Outline a simple scenario for an automated environmental control system (e.g., an automated greenhouse or smart thermostat), explicitly specifying:
- The specific sensor used [1 mark]
- The computer processing threshold condition [1 mark]
- The resulting actuator physical action [1 mark]

5. Processor Design & Instruction Execution
a) Define the core philosophical difference between RISC (Reduced Instruction Set Computer) and CISC (Complex Instruction Set Computer) architectures regarding instruction size and decoding complexity. [3 marks]
b) Explain the role of pipelining in executing instructions concurrently within a RISC processor, and name one hazard that can stall the execution pipeline. [3 marks]
c) State the specific function of the following CPU registers during the instruction cycle: Program Counter (PC), Accumulator (ACC), and Memory Data Register (MDR). [3 marks]

6. Hardware Interrupt Handling
What is a hardware interrupt, and what step-by-step sequence occurs at the machine code level when an interrupt signal is received by the CPU? [4 marks]

7. Digital Logic Circuits & Operations
Draw or describe the logic gate symbols and write out the full truth tables for the three basic logic operations: AND, OR, and NOT. [6 marks]

SELF-ASSESSMENT CHECKLIST

- [ ] Can I explain primary vs. secondary storage?
- [ ] Can I detail RISC vs CISC architectural differences?
- [ ] Can I classify SISD, SIMD, MISD, MIMD systems?
- [ ] Can I describe sensor/actuator ADC/DAC loops?
- [ ] Do I know SRAM (flip-flops) vs DRAM (capacitors)?
- [ ] Do I understand pipelining stages & hazards?
- [ ] Do I know how interrupts alter register state?
- [ ] Can I draw truth tables & simplify logic circuits?